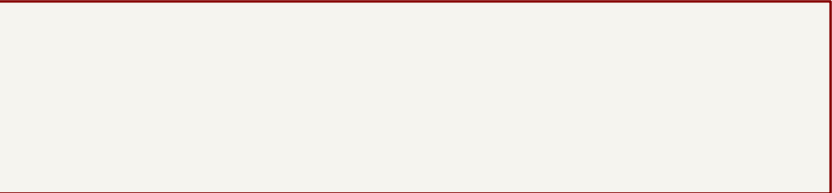


FarmMesh Node / Revision A1 – PROTOTYPE

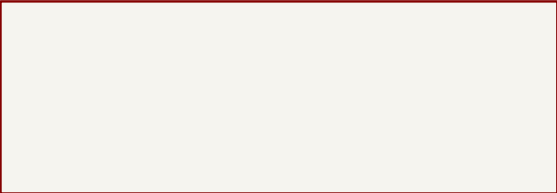
1 x RP2350B + 3 x ESP32-C5-WROOM-1U-N8R8
1S protected Li-ion input; 3.3V regulated rail
Native KiCad 10 schematic + four-layer prototype PCB

1S battery input and 3.3V buck-boost



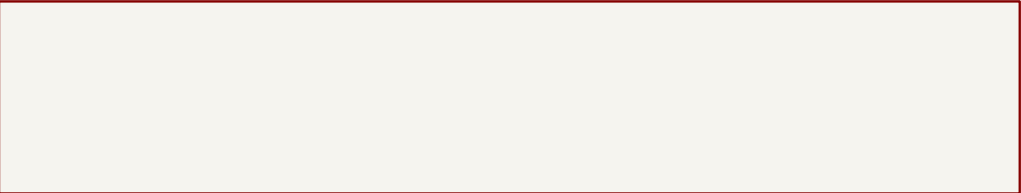
File: 01-power.kicad_sch

Test access and mechanical features



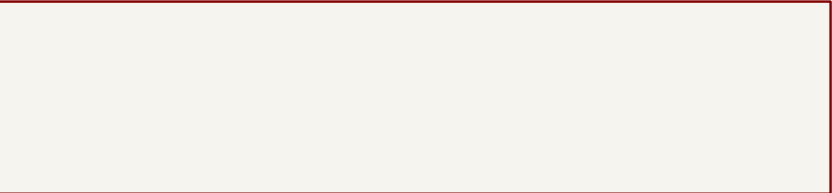
File: 07-test-mechanical.kicad_sch

C5-A module and independent link A



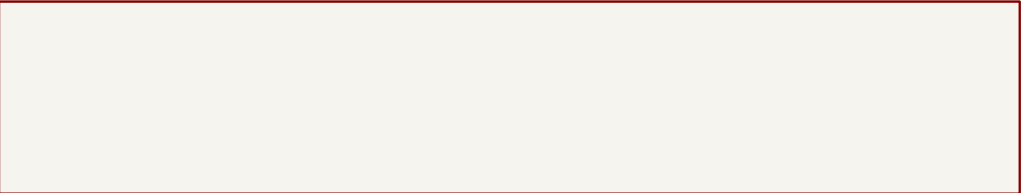
File: 03-radio-a.kicad_sch

RP2350B aggregation, memory and debug



File: 02-rp2350.kicad_sch

C5-B module and independent link B

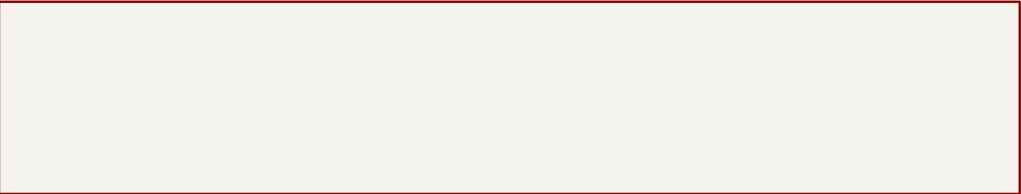


File: 04-radio-b.kicad_sch

INTERFACE CONVENTION

TX / RX are named from the C5 perspective.
Both TX_CLK and RX_CLK originate at that C5.
RP drives only RX_D[0..3] and RX_VALID.
Global net labels make the electrical sheet connections.
No wireless protocol is selected by this schematic.

C5-C module and independent link C

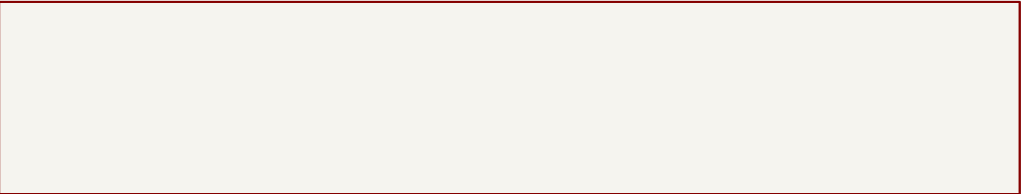


File: 05-radio-c.kicad_sch

REVIEW BOUNDARIES

Battery: conventional 4.20V-max Li-ion, NOT LiFePO4 or HV Li-ion.
Normal loaded operation about 3.0-4.2V; cold start about 3.3V.
External protected pack required. No on-board charger or PV input.
PCB release status: see RELEASE.md and final review reports.
RF isolation, converter stability and throughput require first-board validation.

RP USB boot / self-powered USB-C



File: 06-usb.kicad_sch

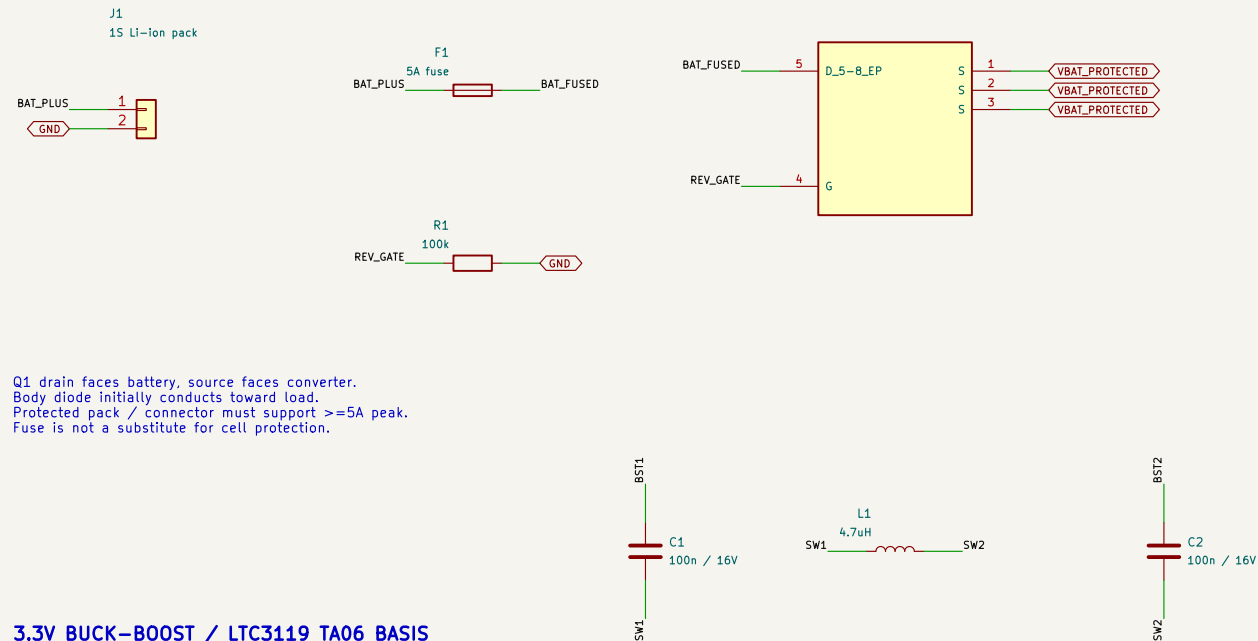
1S Li-ion battery input; no on-board charger
Prototype engineering – release status in RELEASE.md

FarmMesh Node
Sheet: /
File: FarmMesh-Node.kicad_sch

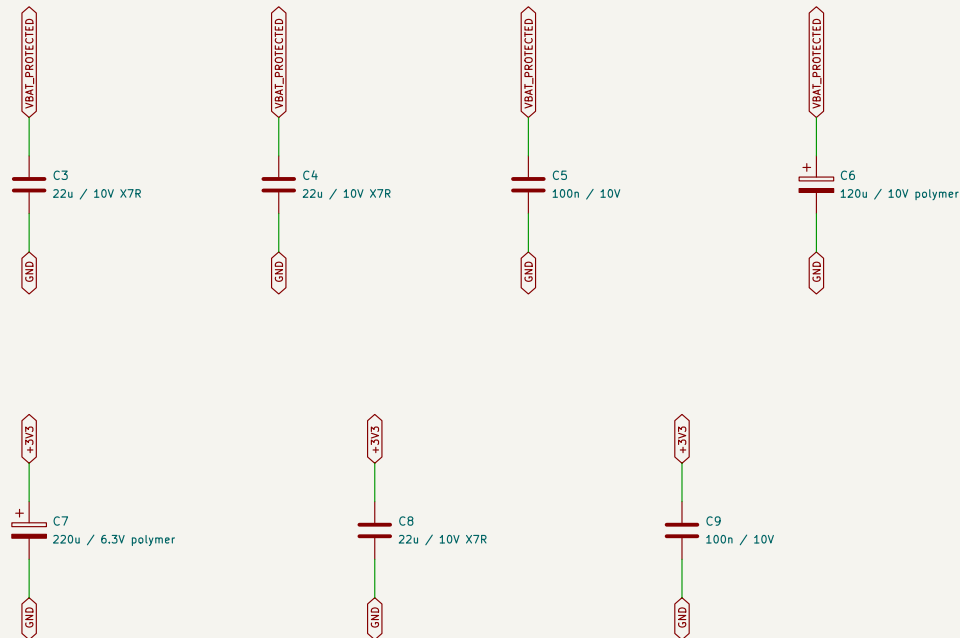
Title: FarmMesh Node – schematic review index

Size: A3	Date: 2026-09-23	Rev: A1
KiCad E.D.A. 10.0.5		Id: 1/8

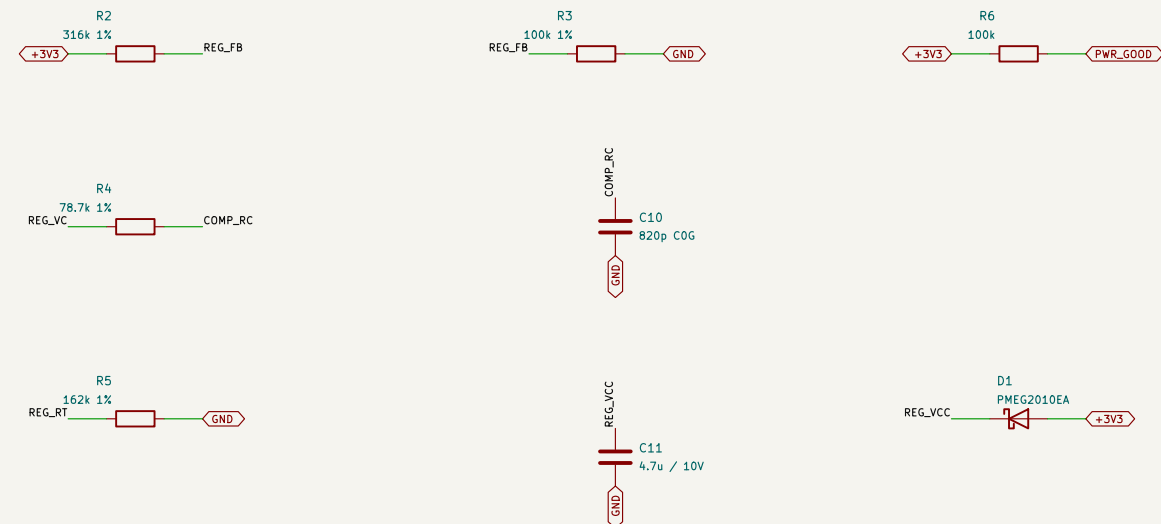
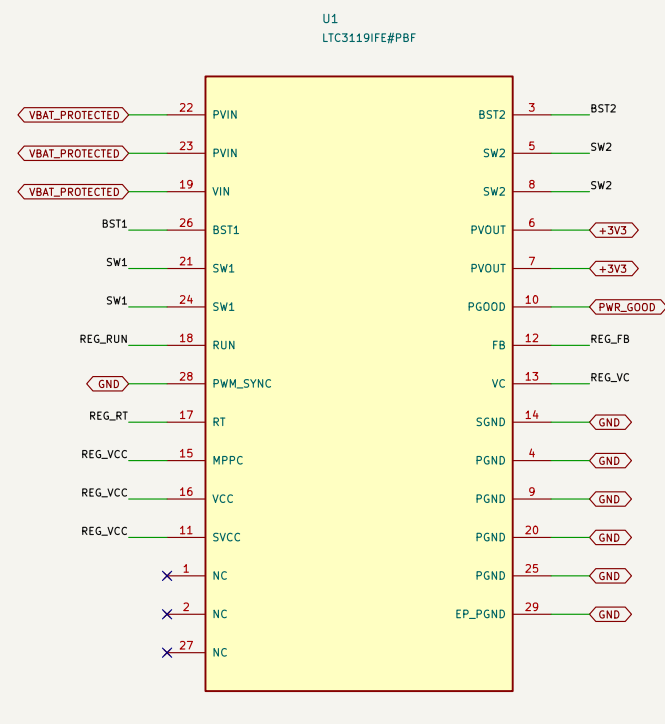
BATTERY INPUT / REVERSE POLARITY



3.3V BUCK-BOOST / LTC3119 TA06 BASIS

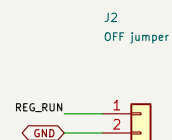
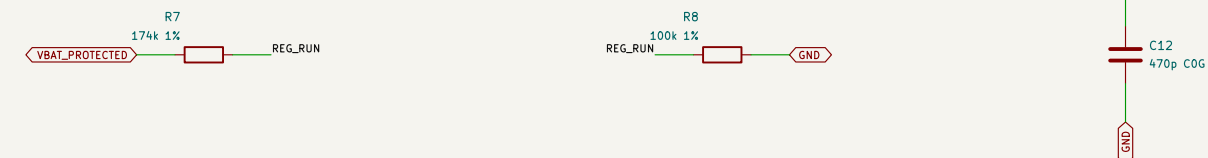


OUTPUT: 2.2A rail budget; 3A transient validation target.
3 x C5 supply capability $\geq 0.6A$ each + RP and margin.
Not a guaranteed continuous 3A rating at every input/temperature.



500kHz; VOUT = $0.795 \times (1 + 316k/100k) = 3.307V$ nominal.
MPPC tied to VCC (disabled). PWM/SYNC = GND (Burst allowed).
SGND and PGND meet at exposed-pad ground; honor TA06 layout.
D1: anode=3V3, cathode=VCC; do not short VCC to 3V3.

BATTERY UVLO / USER OFF



RUN nominal: on 3.30V / off 3.01V; tolerances/load sag apply.
No external charger/PV input is provided.

1S Li-ion battery input; no on-board charger
 Prototype engineering – release status in RELEASE.md
FarmMesh Node
 Sheet: /1S battery input and 3.3V buck-boost/
 File: 01-power.kicad_sch

Title: 1S battery input and 3.3V buck-boost

Size: A3 Date: 2026-09-23

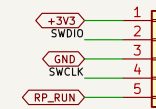
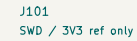
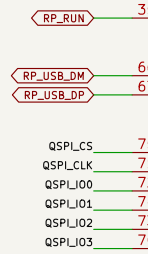
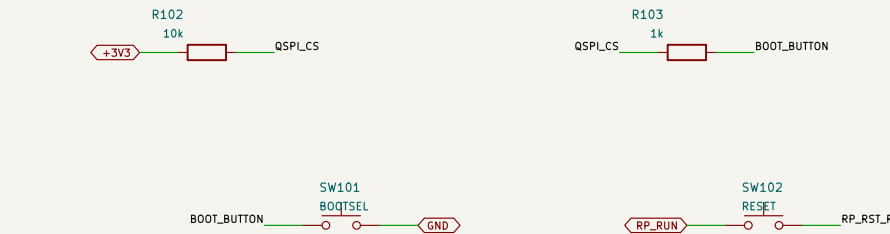
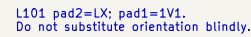
Size: A5	
KiCad E.D.A. 10.0.5	

Date: 2026-09-23

Rev: A1

Id: 2/8

CORE BUCK – follow official R4-S1 placement and L orientation



```

PIO0 BASE0: GPIO0..11 = A
PIO1 BASE0: GPIO12..23 = B
PIO2 BASE16: GPIO24..35 = C
36 = PGOOD; 37 = USB sense (FT pad)
38..47 reserved (NC); battery ADC deferred.
PIO timing/programs NOT validated.

```

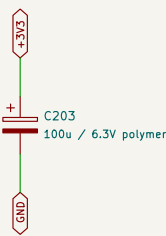
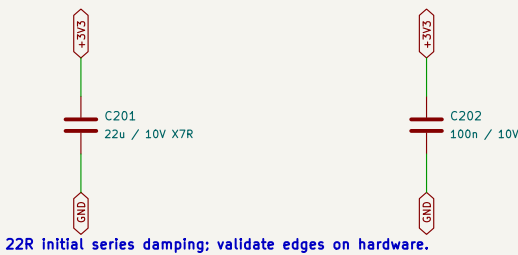
Title: RP2350B aggregation, memory and debug

Rev: A1

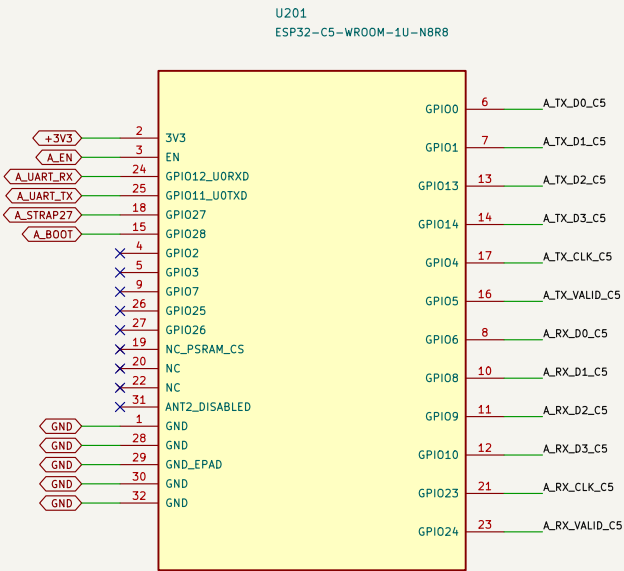
Size: A5	
KiCad E.D.A. 10.0.5	

ESP32-C5 MODULE A / UART PROGRAMMING / INDEPENDENT 12-SIGNAL LINK

C5 TX and RX clocks both originate here. RP never drives either clock. UART0 download retained; C5 native USB disabled to release GPIO13/14. All interface GPIOs avoid boot straps and internal Flash/PSRAM.



Place 22u + 100n at pad2, short ground return. 100u polymer reservoir; measure full rail stability. ANT1 needs an external 2.4/5 GHz antenna and cable. Pad31 ANT2 remains unused on the default module.



C5 source



C5 source



C5 source



C5 source



C5 source



C5 source



RP source



RP source



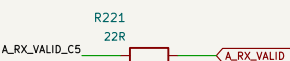
RP source



RP source



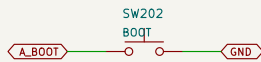
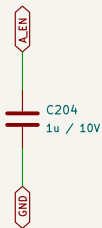
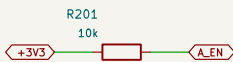
C5 source



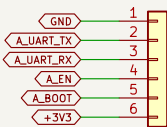
RP source



Place each VALID pull-down at its receiver.



J201
UART / 3V3 reference only



DOWNLOAD: hold BOOT (GPIO28 low), pulse RESET; GPIO27 stays high. 3.3V logic UART only; header pin6 is a voltage reference, not a power input. USB D+ startup transitions may appear on TX data; RP stays input and ignores data until C5 configures PARLIO and asserts VALID. Pre-arm all RX transactions. Series resistors belong at the listed source, regardless of schematic page.

1S Li-ion battery input; no on-board charger
Prototype engineering – release status in RELEASE.md

FarmMesh Node

Sheet: /C5-A module and independent link A/
File: 03-radio-a.kicad_sch

Title: C5-A module and independent link A

Size: A3 Date: 2026-09-23

Rev: A1

KiCad E.D.A. 10.0.5

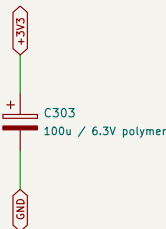
Id: 4/8

ESP32-C5 MODULE B / UART PROGRAMMING / INDEPENDENT 12-SIGNAL LINK

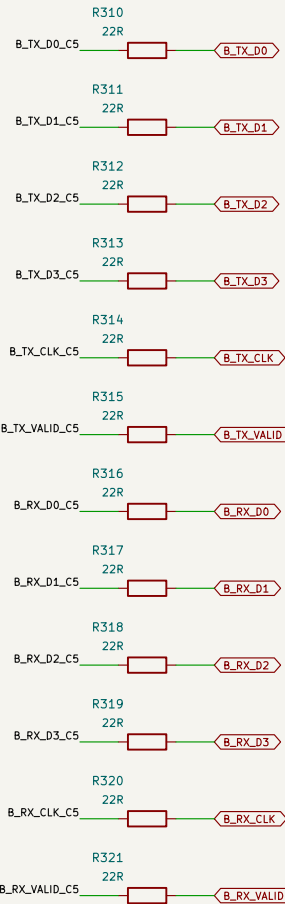
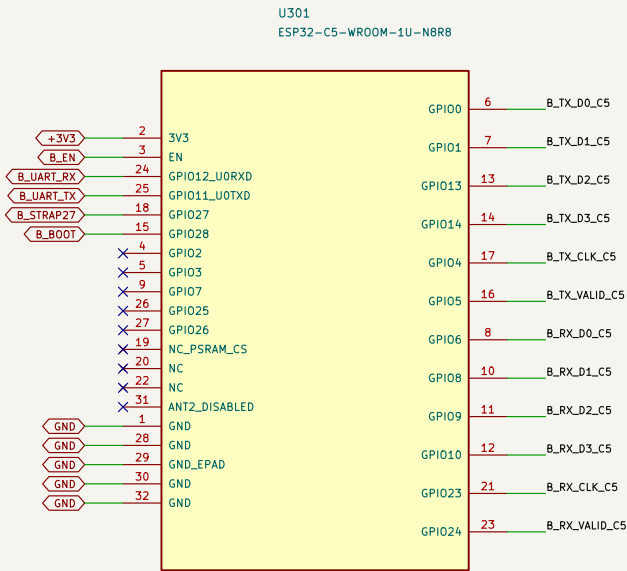
C5 TX and RX clocks both originate here. RP never drives either clock. UART0 download retained; C5 native USB disabled to release GPIO13/14. All interface GPIOs avoid boot straps and internal Flash/PSRAM.



22R initial series damping; validate edges on hardware.



Place 22u + 100n at pad2, short ground return. 100u polymer reservoir; measure full rail stability. ANT1 needs an external 2.4/5 GHz antenna and cable. Pad31 ANT2 remains unused on the default module.



C5 source

C5 source

C5 source

C5 source

C5 source

C5 source

RP source

RP source

RP source

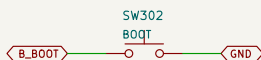
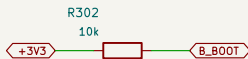
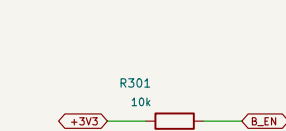
RP source

C5 source

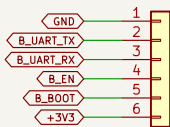
RP source



Place each VALID pull-down at its receiver.



J301
UART / 3V3 reference only



DOWNLOAD: hold BOOT (GPIO28 low), pulse RESET; GPIO27 stays high. 3.3V logic UART only; header pin6 is a voltage reference, not a power input. USB D+ startup transitions may appear on TX data; RP stays input and ignores data until C5 configures PARLIO and asserts VALID. Pre-arm all RX transactions. Series resistors belong at the listed source, regardless of schematic page.

1S Li-ion battery input; no on-board charger
Prototype engineering – release status in RELEASE.md

FarmMesh Node

Sheet: /C5-B module and independent link B/
File: 04-radio-b.kicad_sch

Title: C5-B module and independent link B

Size: A3 Date: 2026-09-23

Rev: A1

KiCad E.D.A. 10.0.5

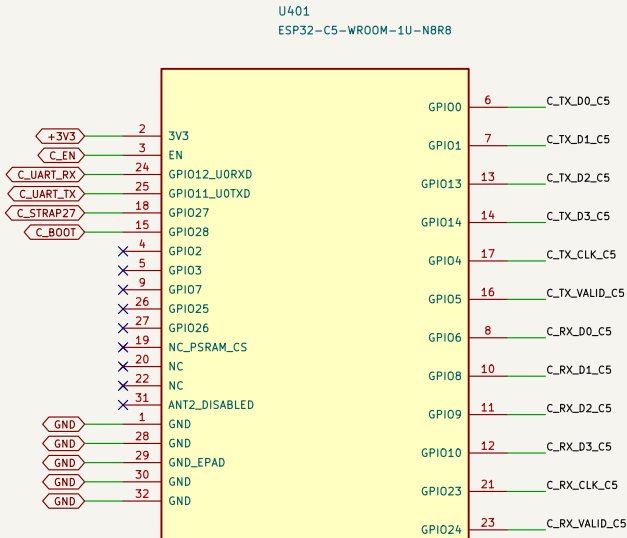
Id: 5/8

ESP32-C5 MODULE C / UART PROGRAMMING / INDEPENDENT 12-SIGNAL LINK

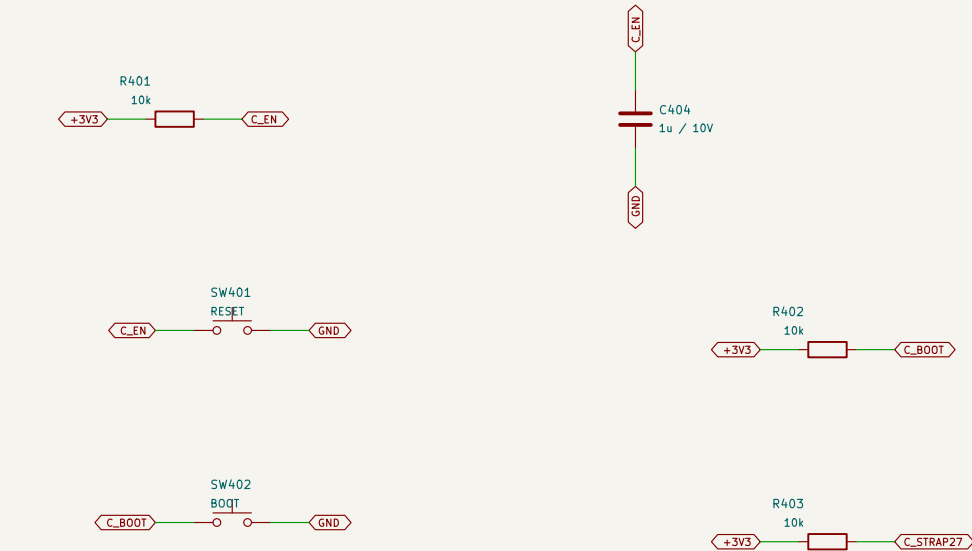
C5 TX and RX clocks both originate here. RP never drives either clock. UART0 download retained; C5 native USB disabled to release GPIO13/14. All interface GPIOs avoid boot straps and internal Flash/PSRAM.



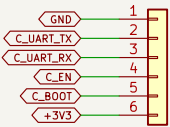
Place 22u + 100n at pad2, short ground return. 100u polymer reservoir; measure full rail stability. ANT1 needs an external 2.4/5 GHz antenna and cable. Pad31 ANT2 remains unused on the default module.



Place each VALID pull-down at its receiver.



J401
UART / 3V3 reference only



DOWNLOAD: hold BOOT (GPIO28 low), pulse RESET; GPIO27 stays high. 3.3V logic UART only; header pin6 is a voltage reference, not a power input. USB D+ startup transitions may appear on TX data; RP stays input and ignores data until C5 configures PARLIO and asserts VALID. Pre-arm all RX transactions. Series resistors belong at the listed source, regardless of schematic page.

1S Li-ion battery input; no on-board charger
Prototype engineering – release status in RELEASE.md
FarmMesh Node
Sheet: /C5-C module and independent link C/
File: 05-radio-c.kicad_sch

Title: C5-C module and independent link C

Size: A3 Date: 2026-09-23

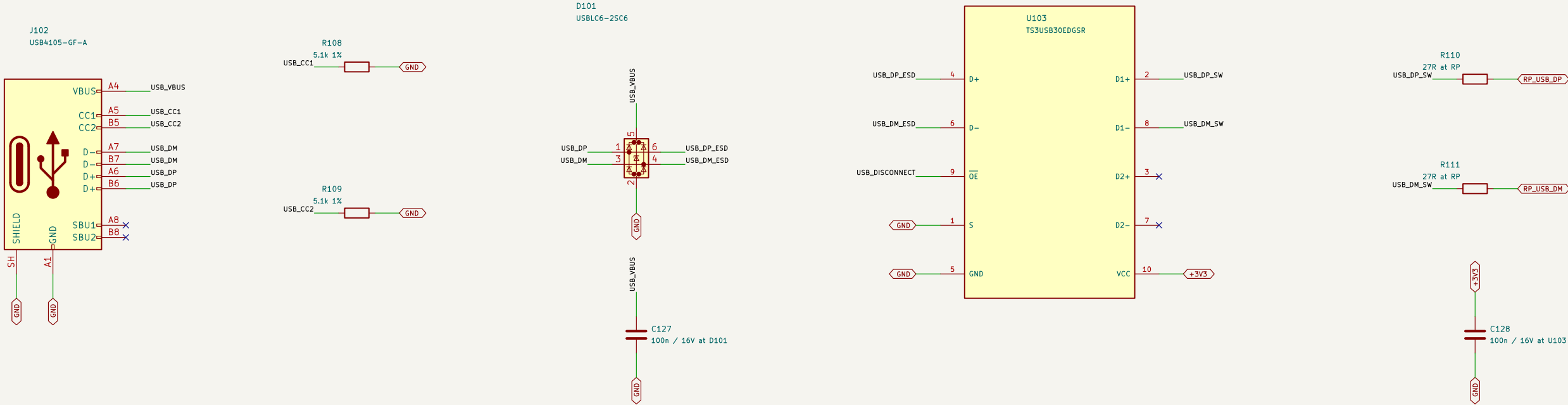
Rev: A1

KiCad E.D.A. 10.0.5

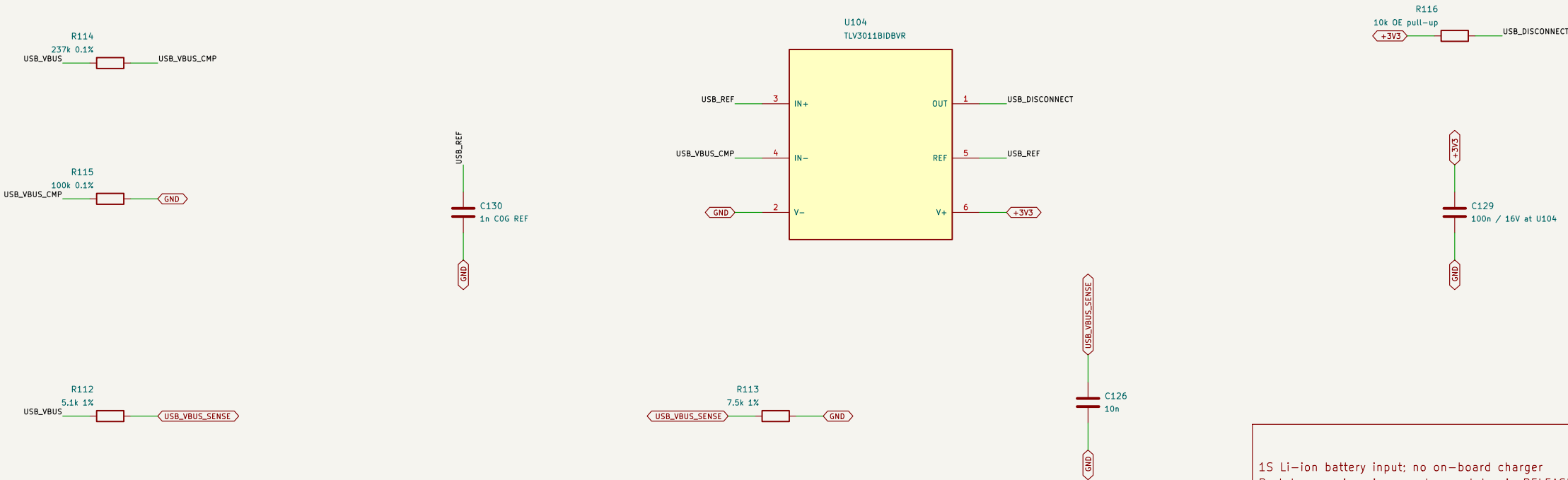
Id: 6/8

RP USB-C / BATTERY POWER REQUIRED

USB VBUS does not power the board or charge the battery.
Hardware disconnects USB data below approximately 4.19V VBUS, including ROM BOOTSEL.
C5 USB pads are allocated to PARLIO; each C5 uses its UART header.



VBUS HARDWARE QUALIFICATION – NO FIRMWARE DEPENDENCY



GPI037: FT input; 5.1k/7.5k divider; disable internal pulls. U104 B variant required.
U103 OE defaults high (disconnected). VBUS trip tolerances/power ordering need sample validation.
90 ohm USB pair: J102 -> D101 -> U103 -> 27R near RP. Keep ESD on connector side.

1S Li-ion battery input; no on-board charger
Prototype engineering – release status in RELEASE.md
FarmMesh Node
Sheet: /RP USB boot / self-powered USB-C/
File: 06-usb.kicad_sch

Title: RP USB boot / self-powered USB-C

Size: A3	Date: 2026-09-23	Rev: A1
KiCad E.D.A. 10.0.5		Id: 7/8

TEST ACCESS / FABRICATED PCB FEATURES

Exposed copper test pads and unplated mounting holes are part of the PCB.
No assembly component is fitted to these references. Test ground first.
All voltages are relative to GND. Use current-limited battery supply on first power-up.

TP1
VBAT_PROTECTED



TP2

+3V3



TP3

GND



TP101

CORE_1V1



TP102

GND



TP103

RP_RUN



TP104

PWR_GOOD



H1

M3 / 3.2mm NPTH



H2

M3 / 3.2mm NPTH



H3

M3 / 3.2mm NPTH



H4

M3 / 3.2mm NPTH



1S Li-ion battery input; no on-board charger
Prototype engineering – release status in RELEASE.md

FarmMesh Node
Sheet: /Test access and mechanical features/
File: 07-test-mechanical.kicad_sch

Title: Test access and mechanical features

Size: A3	Date: 2026-09-23	Rev: A1
KiCad E.D.A. 10.0.5		Id: 8/8